

Intermediate Macroeconomics

BSc. in Business Administration and BSc. in Finance and Accounting

Intermediate Macroeconomics – BMAK

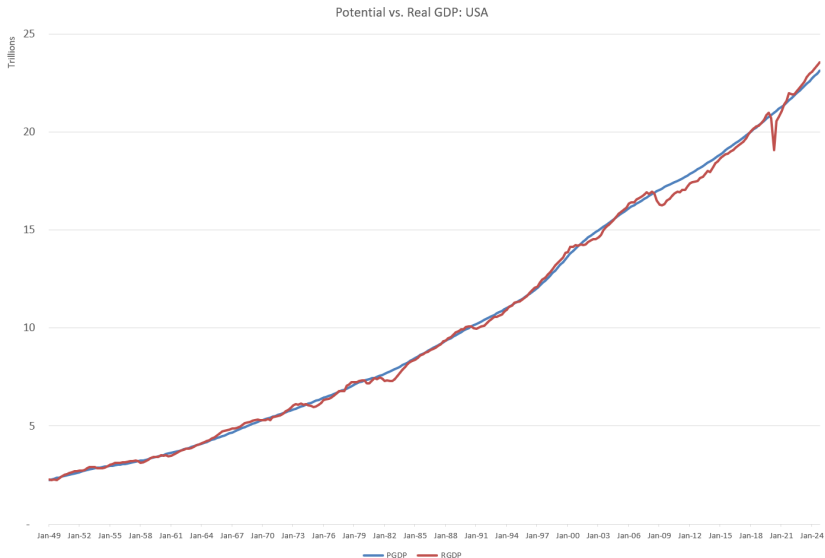
Chapter 4: The Macroeconomy in the Short Run

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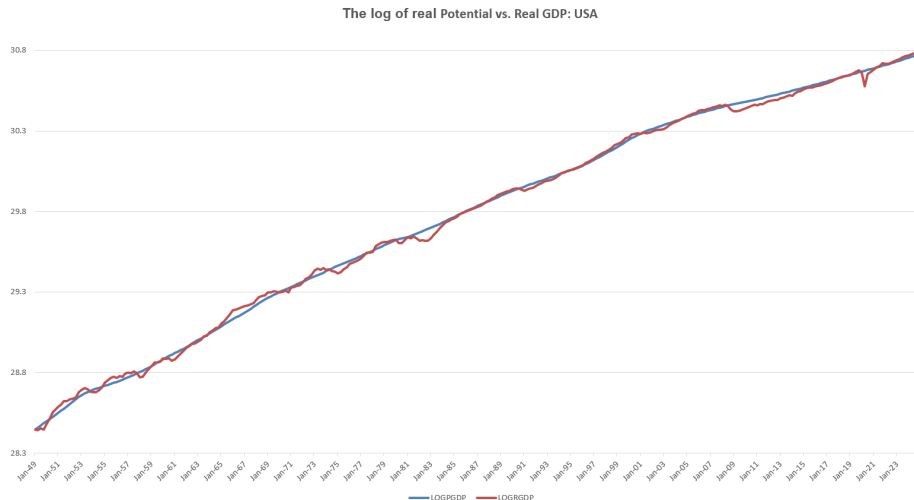
Vietnamese-German University

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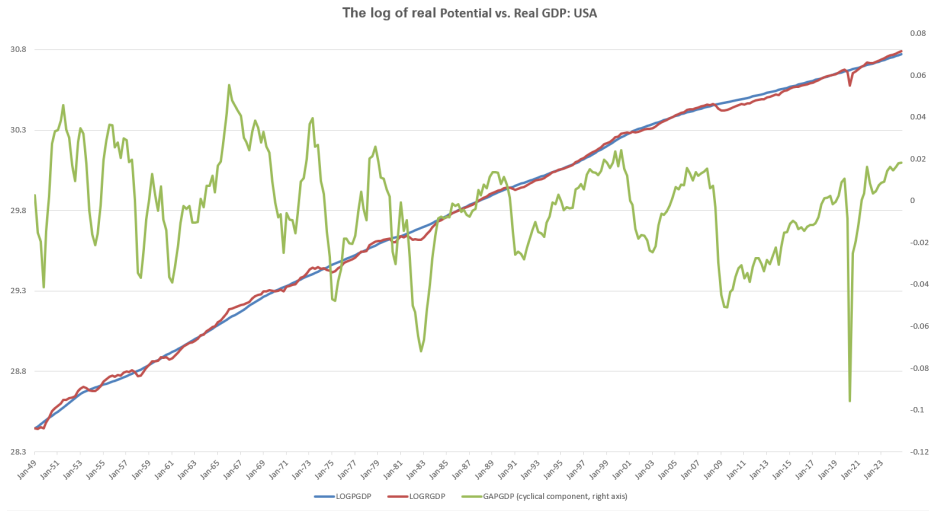
This chart shows the development of real GDP and its potential for the USA in the last 8 decades.



This chart shows the development of real GDP and its potential for the USA in the last 8 decades, in log scale.



This chart shows the development of real GDP and its potential for the USA in the last 8 decades, in log scale.



Two main features stand out:

- (i) **Upward trend**, reflecting positive long-run real GDP growth; in the graphs below, the long-run trend values of (the logarithm of) real GDP have been calculated using filtering techniques from statistics (that are beyond the scope of this course). We will use one technique known as the HP filter during our practice.
- (ii) **Cyclical deviations** between actual real GDP and long-run trend real GDP; the graphs below plot these deviations, specifically the logarithm of actual real GDP minus the long-run trend value of (the logarithm of) real GDP.

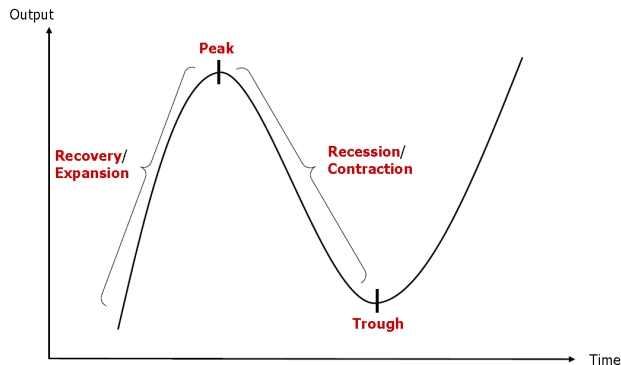
This pattern is called business cycles. A business cycle consists of two phases:

- **Recoveries/expansions**: expanding economic activity that spreads across the economy, normally visible through prolonged positive output growth;
- **Recessions/contractions**: falling economic activity that spreads across the economy, normally visible through prolonged negative output growth.

In stylized form, in a graph with (the logarithm of) output on the vertical axis, a business cycle can thus be represented as follows:

This chart is known as the Burns-Mitchell diagram. It informs us about the typical output dynamics over the course of the business cycle

Graph of a Stylized Business Cycle



Understanding and forecasting business cycles is crucial for:

- **Firms:** to guide production and personnel decisions.
- **Government policymakers:** to stabilize fluctuations, employment, and macroeconomic risk.
- **Financial market investors:** due to the impact of cycles on asset returns and risk.

In this course, we build models explaining business cycles in the **short and medium run**. But first, we must understand **key business cycle characteristics** in the data.

Burns-Mitchell Diagrams

- Illustrate typical output dynamics during business cycles.
- Reveal **co-movement patterns** between output and macro/financial variables.

Two Steps to Construct Burns-Mitchell Diagrams:

- **Identify business cycle peaks:**
 - ▶ A peak = quarter before two consecutive negative growth quarters.
 - ▶ A trough = quarter before two consecutive positive growth quarters.
- **Plot average patterns:**
 - ▶ From 10 quarters before to 10 after each peak.
 - ▶ Plot average (percent deviation from mean) of each variable across all cycles.

To construct the Burns-Mitchell diagram for output (logarithm denoted by Y):

1. Identify:

$$Y_{i,-10}, Y_{i,-9}, \dots, Y_{i,-1}, Y_{i,0}, Y_{i,1}, \dots, Y_{i,10}, \quad i = 1, 2, \dots, M$$

where M is the number of business cycle episodes.

2. Calculate period s averages:

$$\bar{Y}_s = \frac{1}{M} \sum_{i=1}^M Y_{i,s}, \quad s = -10, -9, \dots, 10$$

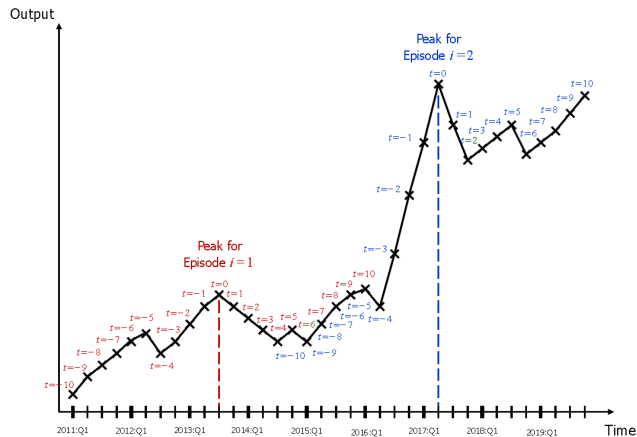
3. Compute the grand average:

$$\bar{Y} = \frac{1}{M} \cdot \frac{1}{21} \sum_{i=1}^M \sum_{s=-10}^{10} Y_{i,s}$$

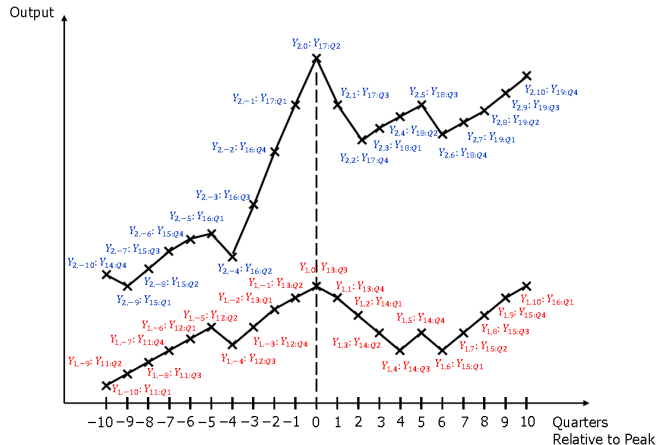
4. Plot:

$$\left(\frac{\bar{Y}_s - \bar{Y}}{\bar{Y}} \right) \cdot 100\%, \quad s = -10, -9, \dots, 10$$

Example: For the sample 2011:Q1 to 2019:Q4, two business cycle episodes are identified.



Example: For the sample 2011:Q1 to 2019:Q4, two business cycle episodes are graphed in a single diagram.



Objective: Measure co-movement of output with other macro/financial variables over the business cycle.
Setup: For two episodes from 2011:Q1 to 2019:Q4, let $C_{i,s}$ denote (log) consumption in episode $i = 1, 2$ and period $s = -10, \dots, 0, \dots, 10$ relative to output-based peak.

Steps (for a general variable X):

- Identify values:

$$X_{i,-10}, X_{i,-9}, \dots, X_{i,0}, \dots, X_{i,10}, \quad i = 1, \dots, M$$

- Compute average at each s :

$$\bar{X}_s = \frac{1}{M} \sum_{i=1}^M X_{i,s}, \quad s = -10, \dots, 10$$

- Compute grand average:

$$\bar{X} = \frac{1}{M \cdot 21} \sum_{i=1}^M \sum_{s=-10}^{10} X_{i,s}$$

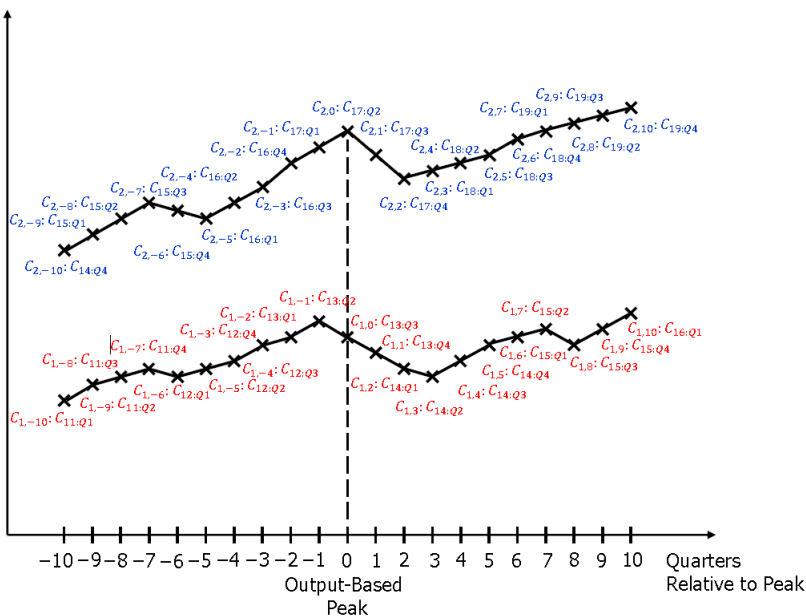
- Plot:

$$\begin{cases} (\bar{X}_s - \bar{X}) \cdot 100\%, & \text{if } X \text{ is in logs} \\ \left(\frac{\bar{X}_s - \bar{X}}{\bar{X}} \right) \cdot 100\%, & \text{otherwise} \end{cases}$$

Terminology:

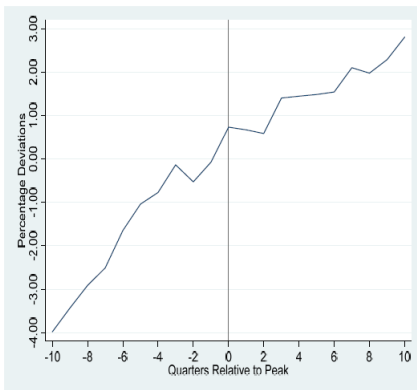
- Pro-cyclical:** Positively correlated with output.
- Counter-cyclical:** Negatively correlated with output.
- Leading:** Peaks/troughs occur *before* those of output.
- Lagging:** Peaks/troughs occur *after* those of output.

Consumption

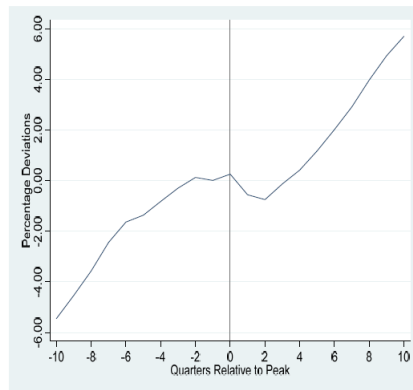


Burns-Mitchell Diagrams for Germany and the U.S.

Real Consumption, Germany
(Quarterly Data, 1970:Q1 to 2020:Q2)



Real Consumption, U.S.
(Quarterly Data, 1970:Q1 to 2020:Q2)

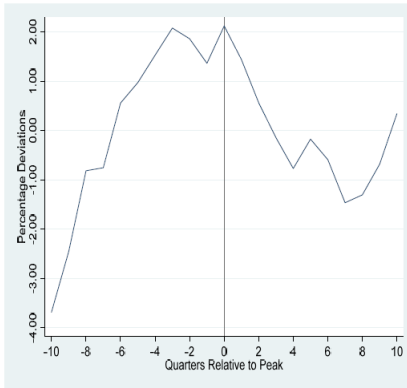


Source of Data: Federal Reserve Bank of St. Louis (2020)

Burns-Mitchell Diagrams for Germany and the U.S.

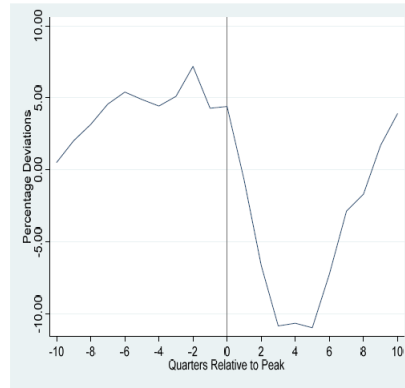
Real Investment, Germany

(Quarterly Data, 1970:Q1 to 2020:Q2)



Real Investment, U.S.

(Quarterly Data, 1970:Q1 to 2020:Q2)

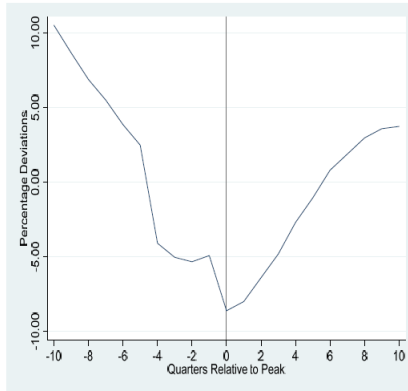


Source of Data: Federal Reserve Bank of St. Louis (2020)

Burns-Mitchell Diagrams for Germany and the U.S.

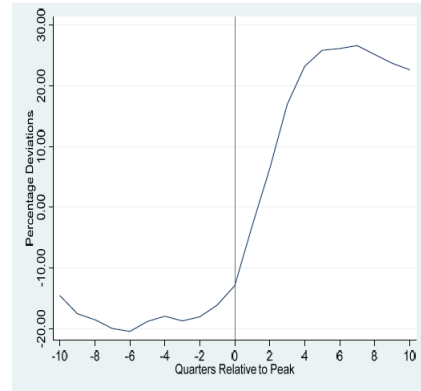
Unemployment, Germany

(Quarterly Data, 1991:Q1 to 2020:Q2)



Unemployment, U.S.

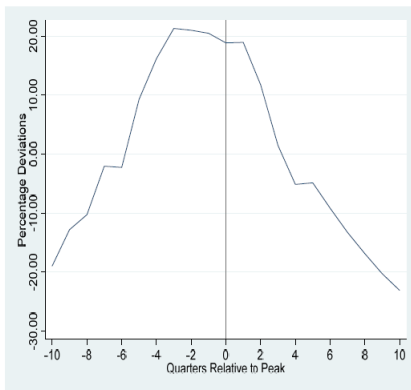
(Quarterly Data, 1970:Q1 to 2020:Q2)



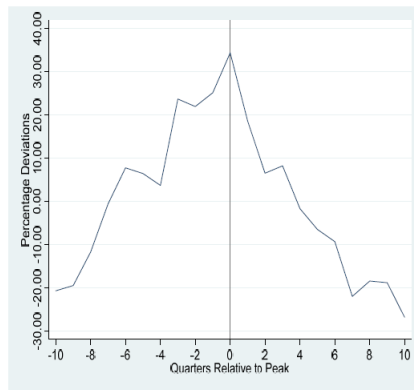
Source of Data: Federal Reserve Bank of St. Louis (2020)

Burns-Mitchell Diagrams for Germany and the U.S.

**Nominal (3 Months)
Interest Rate, Germany**
(Quarterly Data, 1970:Q1 to 2020:Q2)



**Nominal (3 Months)
Interest Rate, U.S.**
(Quarterly Data, 1970:Q1 to 2020:Q2)

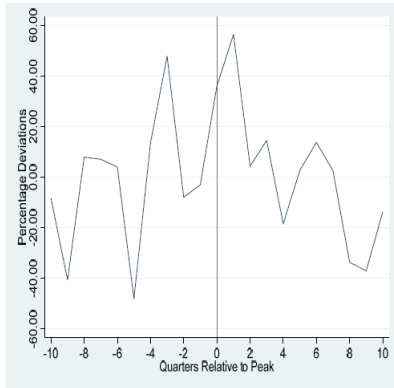


Source of Data: Federal Reserve Bank of St. Louis (2020)

Burns-Mitchell Diagrams for Germany and the U.S.

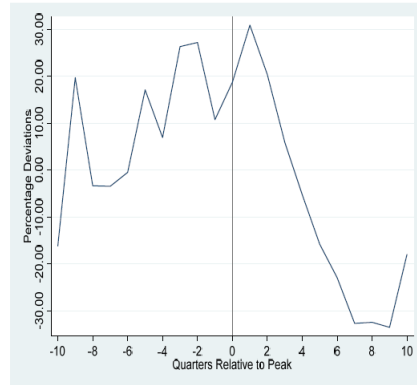
Inflation, Germany

(Quarterly Data, 1970:Q1 to 2020:Q2)



Inflation, U.S.

(Quarterly Data, 1970:Q1 to 2020:Q2)



Source of Data: Federal Reserve Bank of St. Louis (2020)

- Average output loss from peak to trough: about **1% to 2%**.
- Average duration of business cycle:
 - ▶ **2 to 4 quarters** for output.
 - ▶ **6 to 10 quarters** for investment and unemployment.
- **Consumption and investment** are **pro-cyclical**:
 - ▶ Investment is **more volatile** than output.
 - ▶ Consumption is **less volatile** than output.
- **Unemployment** is **counter-cyclical**.
- **(Short-term) nominal interest rates** are **pro-cyclical** (possibly **leading**).
- **Inflation rate** is **pro-cyclical** with **high volatility prior to the peak**.

From the Burns-Mitchell diagrams, we have learned that business cycles involve the systematic co-movement of core macroeconomic aggregates such as output, consumption, investment, (un-) employment and inflation, but also the systematic co-movement between these macroeconomic aggregates and financial market variables (such as interest rates).

Satisfactory models of business cycles will therefore need to capture both the **"real sector"** of the economy (the interaction between the aggregated demand for goods and services from households, firms, government institutions and foreign entities with the production of the goods and services), and the **"financial sector"** of the economy (capturing how interest rates are set through the interaction between savers, borrowers, the central bank and commercial banks).

Developing a model of business cycles will require us to consider the decision makers who contribute to the determination of the key variables in the real and financial sectors. We will therefore:

- analyze the spending decisions of households, firms, government institutions as well as foreign entities, and
- study how financial market participants, specifically the central bank and commercial banks, intermediate and propagate these decisions.

Our models of business cycles will consider business cycles as being initiated by exogenous shocks to components of aggregate demand (such as shocks to the foreign demand for domestically produced goods and services), to financial market variables (such as shocks to risk premia on bonds), or to monetary/fiscal policy rules: **"impulses"**.

These impulses will then be propagated through mechanisms such as feedback loops between aggregate demand and output produced/income generated, and adjustments of aggregate demand due to endogenous changes in financial market conditions: **"propagation mechanisms"**.

We will begin with a **short-run** model of business cycles. The short run is implicitly defined as that time period during which firms' output supply by assumption responds one-to-one to changes in aggregate demand, with **prices** being **constant** ("fixed").

This assumption of fixed prices is a simplifying assumption that we will lift when later turning to a medium-run model of business cycles.

It is worth noting that empirical evidence suggests that for a period of a few quarters:

- Changes in aggregate demand are typically **limited in size** (and thus firms can accommodate these changes through changes only in quantities produced).
- Prices respond **little** to shocks to aggregate demand (outside of episodes of (close to) hyperinflation).

Thus, the **short run** can also be described as a **period of a few quarters**.

In this section, we will study the aggregate demand of the economy and the IS-curve. We begin by modeling the real sector of the economy, analyzing the individual expenditure decisions of households, firms, government institutions as well as foreign entities, and then aggregating these so as to characterize the determinants of the four main components of aggregate demand.

From the expenditure approach of the macroeconomic accounts, we know that these four components are:

- consumption (C),
- investment (I),
- government expenditure (G), and
- trade balance (TB).

Various socio-economic factors influence the level of each component, implying a wide range of potential determinants. However, this analysis will deliberately focus on exploring the relationship between these components and two key policy instruments—namely, the interest rate and autonomous government consumption—which are under the control of the government.

Using the same notation as in our discussion of the macroeconomic accounts, as the baseline case for the **aggregate consumption function**, we will derive in this sub-section the function

$$C = C(\underbrace{Y - T}_{\substack{\text{private sector income, in what} \\ \text{follows called "available} \\ \text{income"} \\ \text{(setting for simplification} \\ BPI = BSI = GT = 0)}}, \underbrace{r}_{\text{interest rate}}, \text{ other factors}), \quad (1)$$

with $\frac{\partial C}{\partial (Y-T)} > 0$ (but less than one), and $\frac{\partial C}{\partial r} < 0$ (but small in absolute terms).

To denote the same facts using more compact notation, we will use

$$C = C(\underbrace{Y - T}_{(+)}, \underbrace{r}_{(-)}, \text{ other factors}). \quad (2)$$

For the quantitative analysis of business cycles, we will require a specific functional form for the aggregate consumption function. We will then use the linear functional form

$$C = \underbrace{C_0}_{\substack{\text{intercept capturing all} \\ \text{factors} \\ \text{other than } Y, T \text{ and } r}} + \underbrace{C_y}_{C_y \in (0,1)} \cdot (Y - T) - \underbrace{C_r}_{C_r > 0} \cdot r, \quad (3)$$

with the parameters C_0 , C_y , and C_r . Considering linear functional forms generally considerably simplifies the algebra when carrying out quantitative analysis.

How can the aggregate consumption functions in Equations (1) to (3) be rationalized as representing household consumption decisions?

When faced with income fluctuations, households for whom the marginal utility from consumption is diminishing wish to **smooth** their **consumption** across time periods.

To see this, consider for simplicity a household living for two periods, which we label the current and the future periods.

The household at the beginning of the current period determines its desired levels of consumption by maximizing utility derived from current consumption and discounted utility derived from future consumption:

$$\max_{C_{\text{current}}, C_{\text{future}}} V = u(C_{\text{current}}) + \left(\frac{1}{1 + \rho} \right) \cdot u(C_{\text{future}}), \quad (4)$$

where

- V denotes lifetime utility,
- $u(\cdot)$ denotes period-by-period instantaneous utility, and
- ρ denotes the parameter measuring how much the household discounts utility from consumption in the future period relative to utility from consumption in the current period, and with the utility maximization being subject to the budget constraints:

$$C_{\text{current}} = \underbrace{(1 + r) \cdot W_{\text{beginning of current period}}}_{\substack{\text{initial wealth at beginning of} \\ \text{current period, plus interest} \\ \text{earned on} \\ \text{it during current period}}} + \underbrace{(Y_{\text{current}} - T_{\text{current}})}_{\substack{\text{current available income} \\ \text{(other than current interest} \\ \text{income)}}} - \underbrace{W_{\text{end of current period}}}_{\substack{\text{wealth set aside at end} \\ \text{of current period for} \\ \text{future consumption}}}, \quad (5)$$

with W denoting real wealth, and

$$C_{\text{future}} = \underbrace{(Y_{\text{future}} - T_{\text{future}})}_{\substack{\text{future available income} \\ \text{(other than future interest} \\ \text{income)}}} + \underbrace{(1 + r) \cdot W_{\text{end of current period}}}_{\substack{\text{initial wealth at end of current} \\ \text{period, plus interest earned on} \\ \text{it during future period}}}. \quad (6)$$

The period-by-period budget constraints in (5) and (6) can be combined into a single lifetime budget constraint: substituting $W_{\text{end of current period}}$ from (6) into (5), we obtain

$$\underbrace{C_{\text{current}} + \frac{C_{\text{future}}}{1+r}}_{\text{PVC (present discounted value of consumption)}} = \underbrace{(1+r) \cdot W_{\text{beginning of current period}} + (Y_{\text{current}} - T_{\text{current}})}_{\text{PVR (present discounted value of resources)}} + \frac{Y_{\text{future}} - T_{\text{future}}}{1+r} \quad (7)$$

Solving the lifetime budget constraint (7) for C_{future} , we obtain

$$C_{\text{future}} = (1+r) \cdot (PVR - C_{\text{current}}). \quad (8)$$

Using Equation (8), we can re-write the household optimization problem (4) in unconstrained form as

$$\max_{C_{\text{current}}} V = u(C_{\text{current}}) + \left(\frac{1}{1+\rho} \right) \cdot u[(1+r) \cdot (PVR - C_{\text{current}})], \quad (9)$$

which in turn implies the first-order condition

$$\frac{\partial V}{\partial C_{\text{current}}} = u'(C_{\text{current}}) - \left(\frac{1}{1+\rho} \right) \cdot \underbrace{u'[(1+r) \cdot (PVR - C_{\text{current}})]}_{=u'(C_{\text{future}})} = 0. \quad (1)$$

$$\Leftrightarrow u'(C_{\text{current}}) = \left(\frac{1+r}{1+\rho} \right) \cdot u'(C_{\text{future}}) \quad (10)$$

(describing that the household at the optimum is indifferent between one more unit of current consumption, and saving that unit and using the proceeds to increase future consumption, a so-called Euler equation)

$$\Leftrightarrow \frac{u'(C_{\text{current}})}{u'(C_{\text{future}})} = \frac{1+r}{1+\rho}. \quad (11)$$

Note that the first-order condition (10) under a diminishing marginal utility from consumption (that is, monotonically decreasing function $u'(\cdot)$) implies a **fixed ratio between current and future consumption** that is **independent of current and future available income**. As an example, consider the instantaneous utility function

$$u(C) = \log(C), \quad (12)$$

with

$$u'(C) = \frac{1}{C} \quad \text{and} \quad u''(C) = -\frac{1}{C^2} < 0,$$

and so marginal utility from consumption is diminishing. Substituting from (12) into (10), we obtain

$$\begin{aligned} \frac{u'(C_{\text{current}})}{u'(C_{\text{future}})} &= \frac{C_{\text{future}}}{C_{\text{current}}} = \frac{1+r}{1+\rho} \\ \Leftrightarrow \frac{C_{\text{current}}}{C_{\text{future}}} &= \frac{1+\rho}{1+r}. \end{aligned} \quad (13)$$

The ratio between current and future consumption is independent of current and future available income (it increases with the rate of discounting, ρ , and decreases with the return on saving, the interest rate r).

How then do households **adjust consumption to changes in their available income**?

Suppose there is a one-time (that is, temporary) increase of $Y_{\text{current}} - T_{\text{current}}$ (note that as business cycles are defined as cyclical deviations from the long-run growth path of income, changes in available income that are part of business cycles are temporary).

From Equation (10), the household will want to

- increase both current *and* future consumption (to maintain a fixed ratio between current and future consumption),

and thus

- increase current consumption by less than the unit increase of $Y_{\text{current}} - T_{\text{current}}$: the **marginal propensity to consume** (*MPC*),

$$\frac{\partial C_{\text{current}}}{\partial (Y_{\text{current}} - T_{\text{current}})} = C_y,$$

must be positive but less than one.

The exact magnitude of the *MPC* will depend on a range of factors, including the possible presence of:

- credit constraints (when current available income is falling, can the household borrow against his/her future available income to fund the smoothing of current consumption?),
- uncertainty about future available income (which can cause additional “saving for a rainy day”), and
- bounded rationality (households being short-sighted).

We will therefore leave it to empirical evidence to pin down the exact magnitude of the **MPC**.

Let us ask next how households **adjust consumption to changes in the interest rate**.

Suppose that there is an increase of the interest rate, r : This leads to both

- a **substitution effect** (making current consumption less attractive as compared to future consumption, as the latter takes advantage of higher returns on savings), and
- an **income effect**:
 - households who before the increase in the interest rate had positive savings ("creditors") now have more income available (and thus can increase current and future consumption), whereas
 - households that before the increase in the interest rate had negative savings ("debtors") now have less income available (and thus have to decrease current and future consumption).
- For creditors, then, the overall effect of an increase in the interest rate on current consumption is ambiguous in sign (depending on the relative strengths of the income and substitution effects).
- For debtors the overall effect of an increase in the interest rate on current consumption is negative in sign (both the substitution and the income effect are negative in sign).
- In the macroeconomy, that features both creditors and debtors, the aggregate effect of an increase in the interest rate on consumption is thus an empirical matter.
- The empirical evidence suggests that following an increase in the interest rate, there is a *small* decrease of aggregate consumption in the macroeconomy.

Summary: The Aggregate Consumption Function

To summarize: We have provided a theoretical rationale for the consumption function in (1),

$$C = C \left(\underbrace{Y - T}_{\text{available income}}, \underbrace{r}_{\text{interest rate}}, \text{other factors} \right), \quad (1)$$

with

$$\frac{\partial C}{\partial (Y - T)} > 0 \quad (\text{but less than one}), \quad \frac{\partial C}{\partial r} < 0 \quad (\text{but small in absolute terms}).$$

We have also seen that the “other factors” influencing (current) consumption include variables such as households’ wealth and their future available income.

Recall once more that **“investment”** in the macroeconomic accounts concerns neither investment in financial assets nor investment in human capital, but rather **accumulation of physical capital**, including purchases of structures and equipment by firms, to be used for the production of goods and services.

As the baseline case for the **aggregate investment function**, we will derive in this sub-section the function

$$I = I \left(\underbrace{r}_{(-)}, \text{ other factors} \right). \quad (14)$$

Thus,

$$\frac{\partial I}{\partial r} < 0,$$

with this effect in absolute value being empirically rather large: *ceteris paribus*, investment is highly sensitive to changes in the rate of interest and thus rather **volatile** over the course of the business cycle.

For the quantitative analysis of business cycles, we will typically specialize the aggregate investment function to

$$I = \underbrace{I_0}_{\substack{\text{intercept capturing all} \\ \text{factors} \\ \text{other than } r}} - \underbrace{I_r}_{>0} \cdot r, \quad (15)$$

with the parameters I_0 and I_r .

How can the aggregate investment functions in Equations (14) and (15) be rationalized as representing the investment decisions of firms?

Consider a two-period model, the two periods denoted as t and $t + 1$, in which a profit-maximizing firm determines its optimal level of investment in period t . Investment in period t increases the period $t + 1$ stock of physical capital, in line with our specification in Chapter II. (see Equation (12) there):

$$K_{t+1} = (1 - \delta) \cdot K_t + I_t.$$

At it is a two-period model, at the end of period $t + 1$ the firm sells that portion of the stock of physical capital that remains after depreciation.

Each period, the firm produces output according to the production function

$$\underbrace{Y_t}_{\text{output}} = F \left(\underbrace{A_t}_{\text{level of technology}}, \underbrace{K_t}_{\text{stock of physical capital}}, \underbrace{L_t}_{\text{labor}} \right). \quad (16)$$

To keep notation as simple as possible, we also assume that the prices of - one unit of physical capital, and of - one unit of output both in periods t and $t + 1$ are equal to one.

The firm then aims to

$$\max_{l_t} \left\{ F(A_t, K_t, L_t) + \underbrace{\left(\frac{1}{1+r} \right)}_{\text{period } t \text{ value of one additional unit of period } t+1 \text{ revenue}} \cdot \underbrace{\left[F(A_{t+1}, K_{t+1}, L_{t+1}) + (1-\delta) \cdot K_{t+1} \right]}_{\text{period } t \text{ value of revenue earned in periods } t \text{ and } t+1} - \left[K_t + l_t + \underbrace{\left(\frac{W_t}{P_t} \right)}_{\text{period } t \text{ real wage}} \cdot L_t + \left(\frac{1}{1+r} \right) \cdot \left(\frac{W_{t+1}}{P_{t+1}} \right) \cdot L_{t+1} \right] \right\}, \quad (17)$$

which in turn implies the first-order condition.

$$\underbrace{\left(\frac{1}{1+r} \right)}_{\text{period } t \text{ value of one additional unit of period } t+1 \text{ revenue}} \cdot \left(\underbrace{\frac{\partial F(A_{t+1}, K_{t+1}, L_{t+1})}{\partial K_{t+1}}}_{=MPK_{t+1}} \cdot \underbrace{\frac{\partial K_{t+1}}{\partial l_t}}_{=1} + (1-\delta) \cdot \underbrace{\frac{\partial K_{t+1}}{\partial l_t}}_{=1} \right) \underbrace{=MR}_{=MR} - \underbrace{\frac{\partial l_t}{\partial l_t}}_{=MC} = 0$$

or, equivalently, that the firm will invest as long as

$$MR = \frac{MPK_{t+1} + 1 - \delta}{1 + r} > 1 = MC$$

$$\Leftrightarrow \frac{MR}{MC} = \text{Tobin's } q = \frac{MPK + 1 - \delta}{1 + r} > 1. \quad (18)$$

$$\frac{\partial(\text{Tobin's } q)}{\partial r} = \frac{\partial [(MPK + 1 - \delta) \cdot (1 + r)^{-1}]}{\partial r} = \frac{-(MPK + 1 - \delta)}{(1 + r)^2} < 0 \quad \Leftrightarrow \quad \frac{\partial I}{\partial r} < 0. \quad (19)$$

Economic reasoning:

- As r is rising, the revenue stream resulting from investment in physical capital (that only accrues in the future) must be discounted at a higher rate.
- For a given cost of investment in physical capital, this makes investment in physical capital less attractive.

Alternative economic reasoning:

- An increase in r increases the opportunity cost of purchasing physical capital: purchasing bonds (or other financial instruments yielding a rate of return r) now yields a higher rate of return, and thus investment in physical capital becomes less attractive.
- As r is rising, firms that are borrowing to purchase additional physical capital face higher costs of borrowing, rendering investment in physical capital less attractive.

Empirically, for publicly traded firms *Tobin's q* is often measured as

$$\text{Tobin's } q = \frac{\text{market value of firm's capital}}{\text{replacement cost of firm's capital}}.$$

If the firm's market value is assessed by financial markets and expectations formed in financial markets are fickle and volatile, then **investment in physical capital will be volatile over the course of the business cycle.**

The magnitude of the sensitivity of investment to the interest rate will depend on a range of factors, including the possible presence of:

- credit constraints (when the interest rate is falling, can the firm borrow against its future revenue to fund current investment expenditure?),
- uncertainty about future revenue (which can create an incentive to wait with investment decisions until more information arrives).

To summarize: We have provided a theoretical rationale for the investment function in (14),

$$I = I \left(\underbrace{r}_{(-)}, \text{ other factors} \right),$$

and have provided reasoning why investment is rather volatile over the course of the business cycle.

We have also seen that the “other factors” influencing (current) investment include firms' production technology and the rate of depreciation of physical capital.

Governments can generally be expected to be **influenced by the political process** as well as **economic considerations**.

Political process:

- Interest groups (within and outside government) may compete and induce the government to "overspend" when the macroeconomy is expanding, leaving government with drained resources in times of recession.

Economic considerations:

- Beyond stimulating aggregate demand in times of recession and saving up in times of expansion ("**countercyclical spending**"), the idea of constraining government spending through "**fiscal rules**" is to ensure debt sustainability, for example by specifying limits for the level of public debt (relative to GDP).

A baseline **aggregate government expenditure function** emphasizing economic considerations is given by:

$$G = G\left(\underbrace{\overline{Y} - Y}_{\substack{\text{long-run} \\ \text{output} \\ \text{(negative of) output} \\ \text{gap} \\ (+)}} ; \underbrace{D}_{\substack{\text{government debt} \\ (-)}}, \text{ other factors} \right). \quad (20)$$

For the quantitative analysis of business cycles, for simplification we will typically specialize the aggregate government expenditure function to explicitly reflect only countercyclical spending:

$$G = \underbrace{G_0}_{\substack{\text{intercept capturing all} \\ \text{factors} \\ \text{other than } \overline{Y} - Y}} + G_y \cdot \underbrace{(\overline{Y} - Y)}_{>0}, \quad (21)$$

with the parameters G_0 and G_y .

Our discussion suggests that G_0 reflects factors such as the political situation and the level of government debt. Also, it should be noted that G_y empirically tends to be rather small.

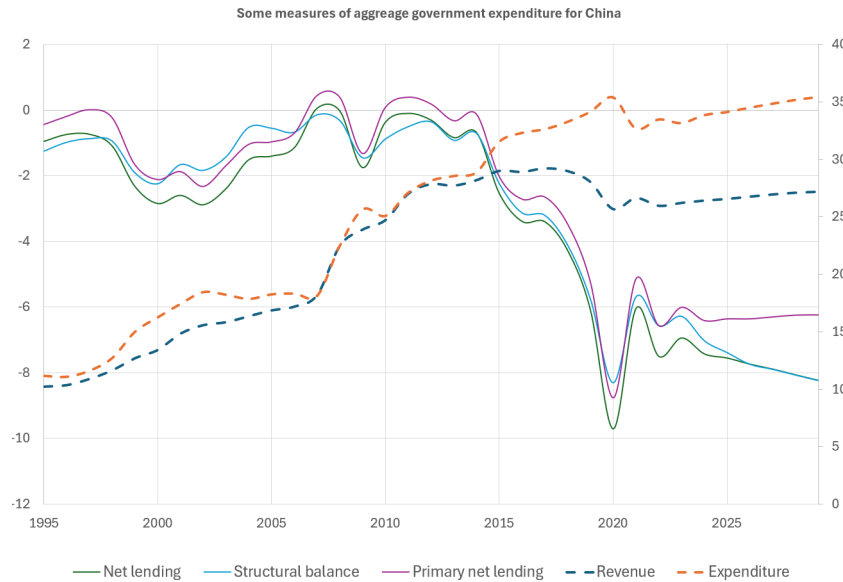
It is important to note that governments may affect output not only directly by controlling G , but also indirectly by influencing C , I and TB , through changes in taxes, transfers and/or expenditure. **Fiscal policy** is the use of government spending and revenue to influence aggregate economic activity, specifically output and employment.

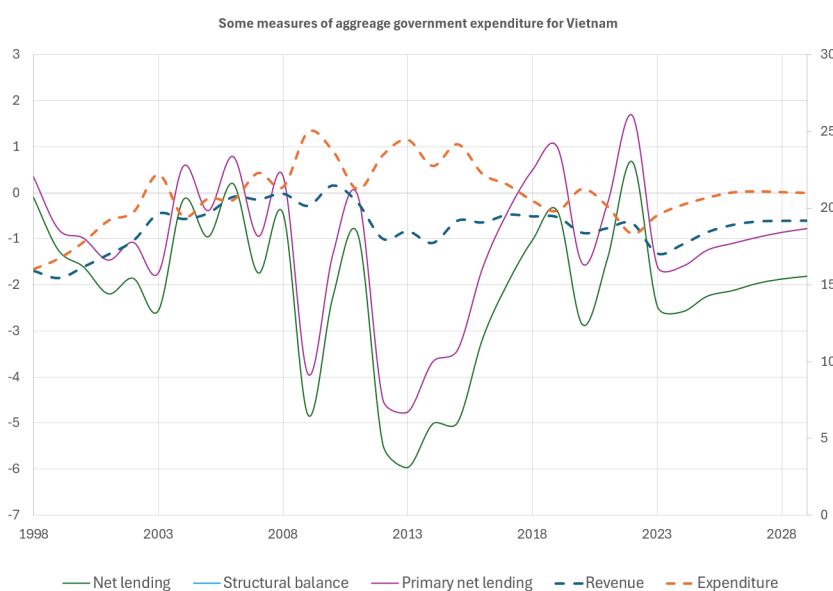
Fiscal policy that increases aggregate demand through an increase in government spending is called **expansionary**. By contrast, fiscal policy is called **contractionary** if it reduces aggregate demand.

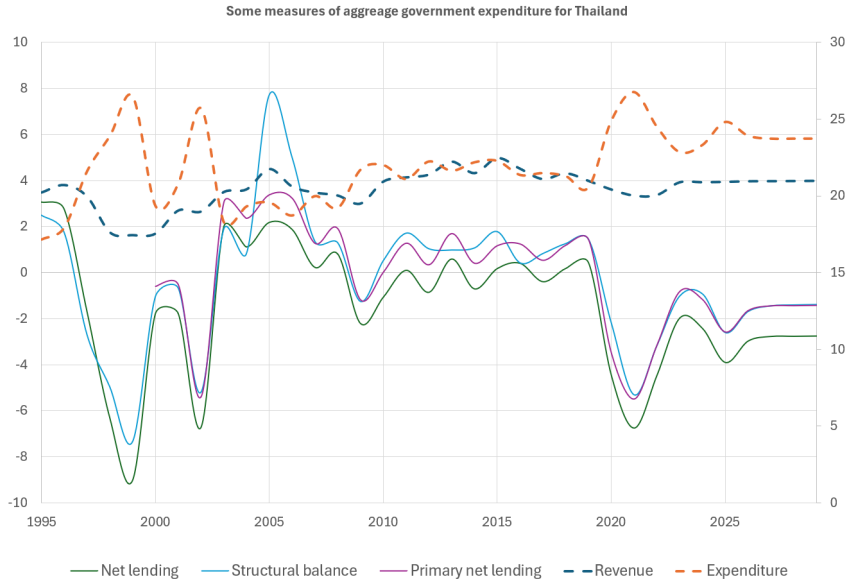
Fiscal policy may affect aggregate demand because of pro-active changes initiated by policymakers, but may also do so even in the absence of such pro-active changes:

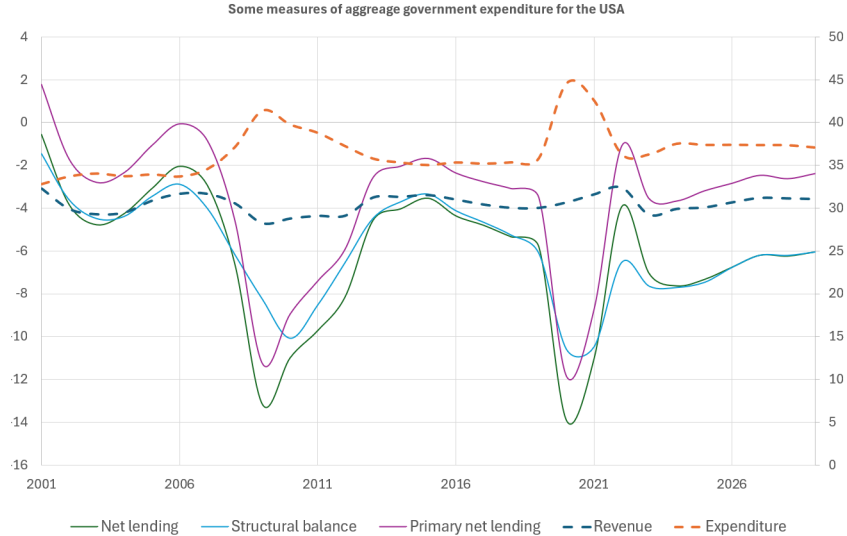
A **fiscal stimulus** entails new discretionary spending or tax cuts (and is thus induced pro-actively).

Also part of fiscal policy are the so-called **automatic stabilizers**: Government tax revenue and expenditure may change without specific action by policymakers. For example, as output falls, for given tax rates less tax revenue is being collected, as corporate profits and households' incomes fall. As another example, government









Let us turn to the fourth main component of aggregate demand, the aggregate trade balance (that is, exports minus imports).

As the baseline case for the aggregate **trade balance function**, we will derive in this sub-section the function

$$TB = TB(\underbrace{Y - T}_{(-)}, \underbrace{Y^* - T^*}_{(+)}, \underbrace{\varepsilon}_{(+)}, \text{ other factors}), \quad (22)$$

where

$$\varepsilon = \varepsilon(\underbrace{r}_{(-)}, \underbrace{r^*}_{(+)}, \text{ other factors}), \quad (23)$$

with starred variables denoting the foreign equivalents of the corresponding domestic variables, and ε denoting the (effective real) exchange rate. As we will elaborate upon in this sub-section, a numerical increase of this exchange rate decreases the price of domestically produced goods and services relative to that of foreign produced goods and services, which in turn increases exports and decreases imports. For the quantitative analysis of business cycles, we will typically specialize the aggregate trade balance and the exchange rate functions (22) and (23) to

$$TB = \underbrace{TB_0}_{\substack{\text{intercept capturing all} \\ \text{factors other than} \\ Y^* - T^*, Y - T, \text{ and } \varepsilon}} + \underbrace{TB_{ex}}_{>0} \cdot (Y^* - T^*) - \underbrace{TB_{im}}_{>0} \cdot (Y - T) + \underbrace{TB_{\varepsilon}}_{>0} \cdot \varepsilon, \quad (24)$$

where

$$\varepsilon = \underbrace{\varepsilon_0}_{\substack{\text{intercept capturing all} \\ \text{factors other than } r \text{ and} \\ r^*}} - \underbrace{\varepsilon_r}_{>0} \cdot (r - r^*), \quad (25)$$

with the parameters TB_0 , TB_{ex} , TB_{im} , TB_{ε} , ε_0 , and ε_r .

How can the aggregate trade balance and exchange rate functions in Equations (22) to (25) be rationalized?

Consider to this purpose first the determination of **imports**:

- **Gross national expenditure** ($GNE = C + I + G$) includes spending both on domestically and foreign produced goods and services. It is sensible to expect that as gross national expenditure increases, so will total imports.
- From our previous discussion, gross national expenditure
 - increases with income: from Equations (3), (15) and (21),

$$\frac{\partial GNE}{\partial(Y - T)} = C_y - G_y \quad \underbrace{\quad}_{\text{empirical evidence}} > 0, \quad (26)$$

- decreases with the rate of interest: from Equations (3), (15) and (21),

$$\frac{\partial GNE}{\partial r} = -C_r - I_r < 0. \quad (27)$$

- The fraction of gross national expenditure on imported (rather than domestically produced) goods and services primarily depends on the **relative price of domestically produced goods and services to that of foreign produced goods and services**, as measured by the effective real exchange rate, ε .

REER vs. NEER

For the discussion to be effective, it is important to have a good understanding of the nominal exchange rate, real exchange rate, nominal effective exchange rate, and the real effective exchange rate.

The bilateral nominal exchange rate measures the rate at which the currencies of the domestic economy, i , and the foreign economy, j , can be exchanged for one another:

$$e_{ij} = \frac{\text{units of domestic currency}}{\text{one unit of foreign currency}}. \quad (28)$$

Note: This is the so-called **price quotation** (which is in line with the quotation of the price for most goods, services and assets, that is, price (in units of domestic currency) per one unit purchased).

Example: Euro Area vs. United States (€/€)

- $e_{\text{€}/\$}$ ↗ ⇔ depreciation of the Euro against the U.S. Dollar
(more Euro needed to purchase one U.S. Dollar);
- $e_{\text{€}/\$}$ ↘ ⇔ appreciation of the Euro against the U.S. Dollar
(fewer Euro needed to purchase one U.S. Dollar).

The bilateral real exchange rate measures the rate at which the goods and services of the domestic economy, i , and the foreign economy, j , can be exchanged for one another:

$$\varepsilon_{ij} = \frac{e_{ij} \cdot P_j}{P_i}, \quad (29)$$

with P_i (P_j) denoting the domestic (foreign) price level, in domestic (foreign) currency units.

REER vs. NEER

Interpreting changes of the bilateral real exchange rate:

- $\varepsilon_{ij} \nearrow \Leftrightarrow$ fall in price of domestically relative to foreign produced goods and services (when expressed in the same currency): competitiveness of domestic economy, i , improving (relative to the foreign economy, j).
- $\varepsilon_{ij} \searrow \Leftrightarrow$ rise in price of domestically relative to foreign produced goods and services (when expressed in the same currency): competitiveness of domestic economy, i , deteriorating (relative to the foreign economy, j).

Finally, the **effective real exchange rate** of the domestic economy, i , is measured by the trade-weighted average of the bilateral real exchange rates between the domestic economy and all the foreign economies, $j = 1, 2, \dots, N$, it does trade with:

$$\varepsilon_i = \sum_{j=1}^N \omega_j \cdot \varepsilon_{ij}, \quad (30)$$

with the weights satisfying the following condition:

$$\sum_{j=1}^N \omega_j = 1. \quad (31)$$

In what follows, we will not require the i subscript on ε_i , and will drop this subscript.

REER vs. NEER

This chart shows the development of the REER and NEER indices for Vietnam in the last 30 years.



Uncovered Interest Parity (UIP)

To complete our discussion of the determination of imports, we will next use the so-called **interest parity relationship** to understand the determination of ε .

In discussing exchange rate determination, we will presume that exchange rates are market-determined (that is, “flexible/floating”), and that there are no restrictions to the cross-border flows of financial capital.

To determine the real exchange rate, consider the following financial investment decision problem: A domestic financial investor seeks to determine at the beginning of period t the optimal portfolio when having the option to invest in either one of the following two assets:

- a (risk-free) domestic-currency-denominated government bond with maturity one period,
- or a (risk-free) foreign-currency-denominated government bond, also with maturity one period.

To keep notation as simple as possible, we also assume that it costs

- one unit of domestic currency to purchase one unit of the domestic-currency-denominated government bond, and
- one unit of foreign currency to purchase one unit of the foreign-currency-denominated government bond.

The domestic financial investor then aims to

$$\max_{B_t, B_t^*} \underbrace{(1 + r_t) \cdot B_t}_{\text{one-period return (in domestic currency) from domestic-currency bond}} + \underbrace{(1 + r_t^*) \cdot B_t^*}_{\text{one-period return (in foreign currency) from foreign-currency bond}} \cdot \underbrace{\varepsilon_{t+1}}_{\substack{\text{domestic currency value of} \\ \text{one unit} \\ \text{of foreign currency in period} \\ t+1}},$$

Uncovered Interest Parity (UIP)

subject to the budget constraint:

$$B_t + \underbrace{\varepsilon_t}_{\substack{\text{domestic currency value of} \\ \text{one unit} \\ \text{of foreign currency in period} \\ t}} \cdot B_t^* = \underbrace{FW_t}_{\substack{\text{funds of financial investor} \\ \text{(in domestic currency)}}},$$

or

$$\max_{B_t} (1 + r_t) \cdot B_t + (1 + r_t^*) \cdot \left(\frac{FW_t - B_t}{\varepsilon_t} \right) \cdot \varepsilon_{t+1}. \quad (32)$$

The optimization problem in (32) implies the first-order condition:

$$\begin{aligned} 1 + r_t + (1 + r_t^*) \cdot \left(-\frac{1}{\varepsilon_t} \right) \cdot \varepsilon_{t+1} &= 0 \\ \Leftrightarrow 1 + r_t &= \underbrace{\frac{1}{\varepsilon_t} \cdot (1 + r_t^*) \cdot \varepsilon_{t+1}}_{\text{interest parity relationship}}, \\ \Leftrightarrow \varepsilon_t &= \left(\frac{1 + r_t^*}{1 + r_t} \right) \cdot \varepsilon_{t+1}. \end{aligned} \quad (33)$$

Equation (33) suggests that exchange rates are driven by domestic relative to foreign interest rates. However, we cannot pin down ε_t from (33) as ε_{t+1} is also unknown. Further analysis is thus needed to determine how ε_t responds to a change in r_t relative to r_t^* .

Uncovered Interest Parity (UIP)

To this purpose, note that interest parity in period $t + 1$ implies that

$$\varepsilon_{t+1} = \left(\frac{1 + r_{t+1}^*}{1 + r_{t+1}} \right) \cdot \varepsilon_{t+2}. \quad (34)$$

Substituting from (34) back into (33), we obtain:

$$\varepsilon_t = \left(\frac{1 + r_t^*}{1 + r_t} \right) \cdot \left(\frac{1 + r_{t+1}^*}{1 + r_{t+1}} \right) \cdot \varepsilon_{t+2}. \quad (35)$$

Using next interest parity in period $t + 2$,

$$\varepsilon_{t+2} = \left(\frac{1 + r_{t+2}^*}{1 + r_{t+2}} \right) \cdot \varepsilon_{t+3}. \quad (36)$$

Upon substituting from (36) back into (35), we obtain:

$$\begin{aligned} \varepsilon_t &= \left(\frac{1 + r_t^*}{1 + r_t} \right) \cdot \left(\frac{1 + r_{t+1}^*}{1 + r_{t+1}} \right) \cdot \left(\frac{1 + r_{t+2}^*}{1 + r_{t+2}} \right) \cdot \varepsilon_{t+3} \\ &= \left[\prod_{s=0}^2 \left(\frac{1 + r_{t+s}^*}{1 + r_{t+s}} \right) \right] \cdot \varepsilon_{t+3}. \end{aligned} \quad (37)$$

Uncovered Interest Parity (UIP)

Repeating these steps to account for interest parity also in periods $t + 3, t + 4, \dots, t + n$, we eventually obtain:

$$\varepsilon_t = \left[\prod_{s=0}^n \left(\frac{1 + r_{t+s}^*}{1 + r_{t+s}} \right) \right] \cdot \varepsilon_{t+n+1}. \quad (38)$$

- Note that for large values of n , ε_{t+n+1} measures the long-term ahead value of the real exchange rate.
- From empirical investigations, we know that in the long run exchange rates tend to be driven by **purchasing power parity**: the nominal exchange rate adapts to ensure that when foreign prices are expressed in domestic currency units, they are equal to domestic prices, that is,

$$P^* \cdot e = P, \quad (39)$$

and thus,

$$\varepsilon = e \cdot P^* / P = 1, \quad (40)$$

stating that in the long run the real exchange rate is constant and equal to one.

Uncovered Interest Parity (UIP)

- **Substituting the result in (40)**, that the long-term ahead value of the real exchange rate is expected to be unity, into (38), we obtain (for large values of n):

$$\varepsilon_t = \left[\prod_{s=0}^n \left(\frac{1 + r_{t+s}^*}{1 + r_{t+s}} \right) \right] = \underbrace{\left(\frac{1 + r_t^*}{1 + r_t} \right)}_{\text{impact of current domestic vs. foreign rate}} \cdot \left[\prod_{s=1}^n \left(\frac{1 + r_{t+s}^*}{1 + r_{t+s}} \right) \right]. \quad (41)$$

- **We thus have the following interest parity implication:** If (starting from an initial interest parity relation) the period t domestic interest rate increases, then in period t the domestic currency appreciates against the foreign currency:

$$\frac{\partial \varepsilon_t}{\partial r_t} = - \left(\frac{1}{1 + r_t} \right)^2 \cdot (1 + r_t^*) \cdot \left[\prod_{s=1}^n \left(\frac{1 + r_{t+s}^*}{1 + r_{t+s}} \right) \right] < 0. \quad (42)$$

Economic reasoning for this interest parity implication:

As the period t interest rate on domestic-currency bonds of maturity one period increases (decreases) (relative to the counterpart interest rate on foreign-currency bonds of maturity one period, and holding constant future interest rates on domestic- and foreign-currency bonds), then

- the demand for domestic-currency bonds increases (decreases),
- leading on the foreign exchange market to an increase (a decrease) of supply of foreign currency and an increase (a decrease) of demand for domestic currency, and
- the current value of the domestic currency appreciates (depreciates).

Uncovered Interest Parity (UIP)

To summarize our discussion of exchange rate determination:

We have provided a theoretical rationale for the exchange rate function in (23),

$$\varepsilon = \varepsilon \left(\underbrace{r}_{(-)}, \underbrace{r^*}_{(+)}, \text{other factors} \right).$$

We have also seen that the "other factors" influencing the current value of the exchange rate include future interest rates and the relative risk of domestic- versus foreign-currency bonds. For the quantitative analysis of business cycles, we will typically specialize the exchange rate function in (23) to (25),

$$\varepsilon = \underbrace{\varepsilon_0}_{\substack{\text{intercept capturing all factors} \\ \text{other than } r \text{ and } r^*}} - \underbrace{\varepsilon_r}_{> 0} \cdot (r - r^*).$$

We will leave it to empirical evidence to pin down the exact magnitudes of the parameters ε_0 and ε_r .

The import function

We can now return to considering the determination of imports. Piecing together our arguments that the value of imports should depend on:

- (i) gross national expenditure, with the latter in turn increasing with domestic disposable income ($Y - T$) and decreasing with the domestic rate of interest (r), as well as
 - (ii) the real exchange rate ($\varepsilon = \varepsilon(r, r^*, \text{other factors})$),
- we have the following aggregate import function:

$$IM = IM[GNE(Y - T, r, \text{other factors}), \varepsilon(r, r^*, \text{other factors})]. \quad (43)$$

Note that

$$\frac{\partial IM}{\partial r} = \underbrace{\frac{\partial IM}{\partial GNE} \cdot \frac{\partial GNE}{\partial r}}_{<0} + \underbrace{\frac{\partial IM}{\partial \varepsilon} \cdot \frac{\partial \varepsilon}{\partial r}}_{>0} \quad (44)$$

$\underbrace{\hspace{15em}}_{\text{imports becoming more expensive for domestic economy as domestic currency depreciates}}$

Based on empirical evidence, we henceforth assume: $\frac{\partial IM}{\partial r} > 0$, (45)
 and can thus write the aggregate import function as

$$IM = IM(\underbrace{Y - T}_{(+)}, \underbrace{\varepsilon}_{(-)}, \text{other factors}), \quad (46)$$

where $\varepsilon = \varepsilon(\underbrace{r}_{(-)}, \underbrace{r^*}_{(+)}, \text{other factors})$

The export function

Moving to exports, as they are the imports of the rest of the world, they are driven by the same set of factors as the domestic imports, but from the foreign perspective. We thus have the aggregate export function:

$$EX = EX(\underbrace{Y^* - T^*}_{(+)}, \underbrace{\varepsilon}_{(+)}, \text{other factors}), \quad (47)$$

exports of domestic economy
becoming less expensive for
rest of the world as domestic
currency depreciates

where

$$\varepsilon = \varepsilon(\underbrace{r}_{(-)}, \underbrace{r^*}_{(+)}, \text{other factors}).$$

Combining the aggregate import and export functions in (46) and (47), we obtain the aggregate trade balance function specified in (22) and (23):

$$\begin{aligned} TB &= EX(\underbrace{Y^* - T^*}_{(+)}, \underbrace{\varepsilon}_{(+)}, \text{other factors}) - IM(\underbrace{Y - T}_{(+)}, \underbrace{\varepsilon}_{(-)}, \text{other factors}) \\ &= TB(\underbrace{Y - T}_{(-)}, \underbrace{Y^* - T^*}_{(+)}, \underbrace{\varepsilon}_{(+)}, \text{other factors}), \end{aligned}$$

where

$$\varepsilon = \varepsilon(\underbrace{r}_{(-)}, \underbrace{r^*}_{(+)}, \text{other factors}).$$

To summarize: We have provided a theoretical rationale for the aggregate trade balance function in (22) and (23).

We have also seen that the “other factors” influencing the aggregate trade balance include

- all “other factors” affecting gross national expenditure (such as households’ wealth and their future disposable incomes, firms’ production technology and the rate of depreciation of physical capital, as well as the political situation and the level of government debt), and
- all “other factors” influencing the current value of the exchange rate (such as future interest rates and the relative risk of domestic- versus foreign-currency bonds).

The *IS*-curve ("*IS*" denoting investment equal to saving) renders for all possible levels of the interest rate the level of output at which **output supply is equal to aggregate demand**. The *IS*-curve together with a relationship determining the interest rate, inter alia as a function of the level of output (this relationship will come from the financial sector), will later allow us to obtain the overall short-run macroeconomic outcomes for output, the interest rate, the components of aggregate demand and the exchange rate. Let us derive and analyze the *IS*-curve. Setting output produced equal to desired aggregate demand, we have:

$$\begin{aligned}
 \underbrace{Y}_{\text{output produced}} &= \underbrace{DD}_{\text{desired aggregate demand}} \\
 &= \underbrace{C_0 + C_y \cdot (Y - T) - C_r \cdot r}_{= C, \text{ from (3)}} \\
 &\quad + \underbrace{I_0 - I_r \cdot r}_{= I, \text{ from (15)}} + \underbrace{G_0 + G_y \cdot (\bar{Y} - Y)}_{= G, \text{ from (21)}} \\
 &\quad + \underbrace{TB_0 + TB_{\text{ex}} \cdot (Y^* - T^*) - TB_{\text{im}} \cdot (Y - T) + TB_{\varepsilon} \cdot [\underbrace{\varepsilon_0 - \varepsilon_r \cdot (r - r^*)}_{= \varepsilon, \text{ from (25)}}]}_{= TB, \text{ from (24)}}
 \end{aligned} \tag{48}$$

Note that in (48), the **endogenous** variables (determined model-internally as functions of the **exogenous** variables and the model parameters) are:

- Y (to be determined such that output supply is equal to desired aggregate demand),
- r (variable that from the Burns-Mitchell diagrams systematically co-varies with output – and thus must be endogenous, too),
- C, I, G, TB, ε (variables that depend on Y and/or r).

The variables that we will keep as exogenous (being determined outside the model), are: $T, \bar{Y}, Y^*, T^*, r^*$. We thus

- specify the domestic tax revenue (T) as "lump-sum" (that is, as fixed, independent of the level of income),
- consider potential output ($\bar{Y}$) as not affected by short-run macroeconomic outcomes, and
- abstract from influences of the domestic economy on foreign variables (Y^*, T^*, r^*): we think of the domestic economy as being "small" relative to the rest of the world economy.

To move towards solving for the endogenous variables, let us split desired aggregate demand into three sub-components:

- a component that is exogenous (also called "autonomous"),
- a component that is income-sensitive, and
- a component that is interest-rate-sensitive:

$$\begin{aligned}
 Y = DD = & \underbrace{C_0 + I_0 + G_0 + TB_0 + TB_\varepsilon \cdot \varepsilon_0 - (C_y - TB_{im}) \cdot T + G_y \cdot \bar{Y} + TB_{ex} \cdot (Y^* - T^*) + TB_\varepsilon \cdot \varepsilon_r \cdot r^*}_{\text{A (autonomous component of desired aggregate demand)}} \\
 & - \underbrace{(C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot r}_{\text{interest rate-sensitive component of desired aggregate demand}} + \underbrace{(C_y - TB_{im} - G_y) \cdot Y}_{\text{income-sensitive component of desired aggregate demand}}
 \end{aligned} \quad (49)$$

Solving (49) for Y , we have

$$[1 - (C_y - TB_{im} - G_y)] \cdot Y = A - (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot r, \quad (50)$$

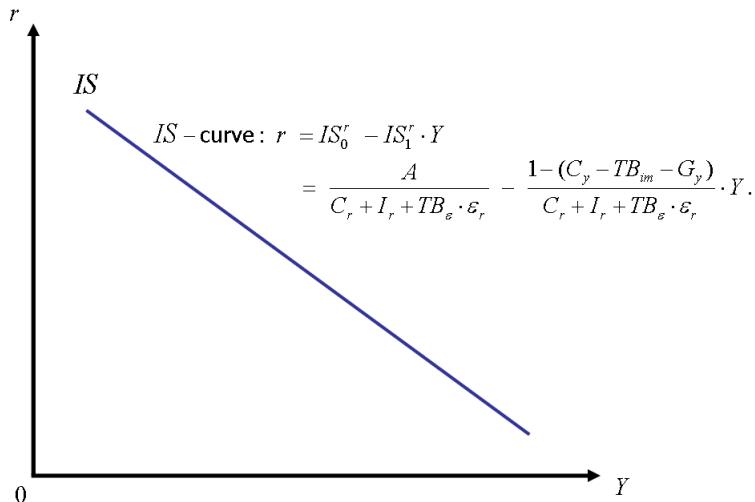
and thus

$$Y = \underbrace{\frac{A}{1 - (C_y - TB_{im} - G_y)}}_{= IS_0} - \underbrace{\frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{1 - (C_y - TB_{im} - G_y)}}_{= IS_1} \cdot r, \quad (51)$$

The relationship in Equation (51) is called the **IS-curve**. Graphically, the convention is to plot the *IS*-curve in a graph with r on the vertical axis and Y on the horizontal axis. We thus re-arrange (51) as follows:

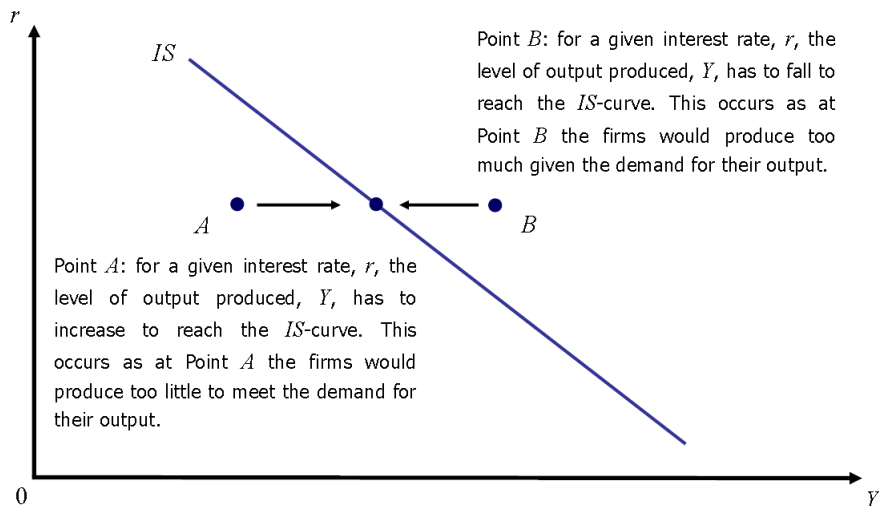
$$\begin{aligned}
 Y &= IS_0 - IS_1 \cdot r \Leftrightarrow IS_1 \cdot r = IS_0 - Y, \\
 \Leftrightarrow r &= \frac{IS_0}{IS_1} - \underbrace{\frac{1}{IS_1}}_{= \frac{1}{IS_1'}} \cdot Y = \underbrace{\frac{A}{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}}_{= IS_0'} - \underbrace{\frac{1 - (C_y - TB_{im} - G_y)}{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}}_{= IS_1'} \cdot Y.
 \end{aligned} \quad (52)$$

We thus have the famous IS curve:



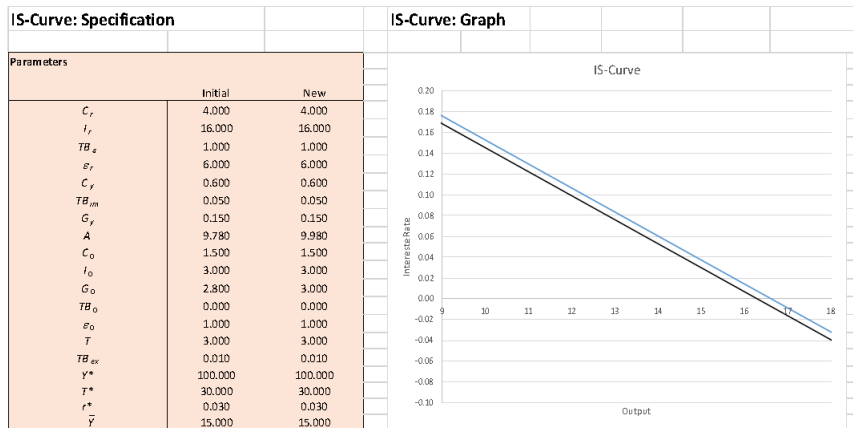
The economy must settle on the IS curve, given each interest rate level.

What happens off the *IS*-curve? Output supply is not equal to desired aggregate demand:



Workbook for the *IS*-Curve

The workbook *IS-Curve.xlsm* allows to study all the determinants of the position and slope of the *IS*-curve.



The IS -curve suggests **two main sources of changes in short-run output**:

- changes in the autonomous component of aggregate demand, A ,
- changes in the interest rate, r .

Let us first turn to discussing **changes in the autonomous component of aggregate demand, A** :

Suppose there is an increase in A by ΔG_0 , that is, government expenditure is rising exogenously by ΔG_0 .

Note from the IS -curve (51) that

$$\frac{\partial Y}{\partial A} \cdot \Delta G_0 = \frac{1}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta G_0. \quad (53)$$

Note that $C_y - TB_{im} - G_y \in (0, 1)$, as

- the marginal propensity to consume, C_y , is less than one,
- the demand for imported consumption goods and services following an increase in disposable income rises only by a fraction of the overall increase in consumption expenditure (that is, some of the increase in the demand for consumption goods and services will be on domestically produced consumption goods and services, $C_y - TB_{im} > 0$),
- G_y empirically tends to be rather small.

It follows that

$$KM = \underbrace{\frac{1}{1 - (C_y - TB_{im} - G_y)}}_{\substack{\in (0, 1) \\ > 1}} \cdot \quad (54)$$

Keynesian multiplier

To understand this Keynesian multiplier, note that an increase in A by ΔG_0 has the following effects:

- From (48), output immediately increases by ΔG_0 (“first-round increase”: as prices are fixed in the short run, firms increase production one-to-one with the initial increase in desired aggregate demand).
- But if Y increases by ΔG_0 , then desired aggregate demand from (49) in turn further increases by $(C_y - TB_{im} - G_y) \cdot \Delta G_0$. This in turn implies a further (“second-round”) increase of Y by $(C_y - TB_{im} - G_y) \cdot \Delta G_0$ (from (48), firms increasing production one-to-one with the further increase in desired aggregate demand).
- The second-round increase of Y leads to another increase of desired aggregate demand, from (49) by the magnitude of

$$(C_y - TB_{im} - G_y) \cdot \underbrace{(C_y - TB_{im} - G_y) \cdot \Delta G_0}_{\text{second-round effect}} = (C_y - TB_{im} - G_y)^2 \cdot \Delta G_0.$$

Thus, from (48), firms increase production one-to-one with this further increase in desired aggregate demand, and there is a “third-round” increase of Y by

$$(C_y - TB_{im} - G_y)^2 \cdot \Delta G_0.$$

- ... and so forth ...

The total increase in Y following the increase of G_0 by ΔG_0 is therefore given by

$$\begin{aligned} \Delta Y = & \underbrace{\Delta G_0}_{\text{first-round effect}} + \underbrace{(C_y - TB_{im} - G_y) \cdot \Delta G_0}_{\text{second-round effect}} + \underbrace{(C_y - TB_{im} - G_y)^2 \cdot \Delta G_0}_{\text{third-round effect}} \\ & + \underbrace{(C_y - TB_{im} - G_y)^3 \cdot \Delta G_0}_{\text{fourth-round effect}} + \dots \end{aligned}$$

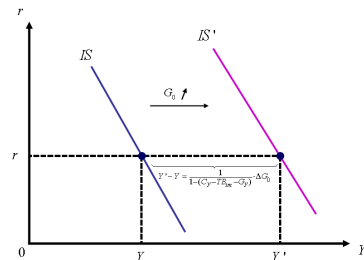
$$\begin{aligned}\Delta Y &= \left[\sum_{t=0}^{\infty} (C_y - TB_{im} - G_y)^t \right] \cdot \Delta G_0 \\ &= \frac{1}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta G_0,\end{aligned}\quad (55)$$

with

$$\frac{1}{1 - (C_y - TB_{im} - G_y)}$$

being the Keynesian multiplier in Equation (54).

Graphically, we can depict the effects of an increase in A by ΔG_0 as follows:



To complete discussion of the IS -curve, let us turn to examining how **changes in the interest rate** affect output. (Note that unlike the autonomous component of aggregate demand, A , which is exogenous, the interest rate, r , is endogenous. We postpone discussion as to how changes in r can arise due to monetary policy decisions to the next section. For now, our interest is only in how a given change in r affects Y .)

Graphically, the output effects of changes in r clearly are captured through movements along the IS -curve. Analytically, from the IS -curve (51) it follows that

$$\frac{\partial Y}{\partial r} \cdot \Delta r = \frac{-(C_r + I_r + TB_\varepsilon \cdot \varepsilon_r)}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta r. \quad (56)$$

Let us first turn to the numerator on the right-hand side of (56):

- This numerator (in $-C_r - I_r$) captures that as the interest rate, r , increases, consumption and investment expenditures decrease, reflecting the higher cost of borrowing ("**consumption and investment channels**" of the response of output to changes in the interest rate).
- This numerator also captures (in $-TB_\varepsilon \cdot \varepsilon_r$) that as the interest rate, r , increases, we also observe an appreciation of the current value of the domestic currency, which in turn decreases exports and increases imports ("**exchange rate channel**" of the response of output to changes in the interest rate).

To understand the total magnitude with which a change in r affects Y , we further need to take into account the denominator on the right-hand side of (56). What explains its magnitude?

- Following the increase of the interest rate by Δr , due to the consumption, investment and ex-change rate channels there is an initial ("first round") change of output, Y , by $-(C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot \Delta r$ (firms decreasing production one-to-one with the interest rate-implied decrease in desired demand).

- But if output, Y , changes by $-(C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot \Delta r$, then desired aggregate demand from (49) in turn further changes by

$$-(C_y - TB_{im} - G_y) \cdot (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot \Delta r$$

(desired aggregate demand falling as Y decreases). This in turn implies a further (“second-round”) change of Y by

$$-(C_y - TB_{im} - G_y) \cdot (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot \Delta r$$

(from (48), firms decreasing production one-to-one with the further decrease in desired aggregate demand).

- The second-round change of Y leads to another change of desired aggregate demand, from (49) by the magnitude of

$$-(C_y - TB_{im} - G_y) \cdot \underbrace{(C_y - TB_{im} - G_y) \cdot (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot \Delta r}_{\text{second-round effect}}$$

$$= -(C_y - TB_{im} - G_y)^2 \cdot (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot \Delta r$$

(desired aggregate demand falling again as Y decreases again). Thus, from (48), firms decrease production again one-to-one with this further decrease in desired aggregate demand, and there is a “third-round” change of Y by

$$-(C_y - TB_{im} - G_y)^2 \cdot (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot \Delta r.$$

- ... and so forth ...

The total change in Y following the increase of r by Δr is therefore given by

$$\begin{aligned}
 \Delta Y &= \underbrace{-(C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot \Delta r}_{\text{first-round effect}} - \underbrace{(C_y - TB_{im} - G_y) \cdot (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot \Delta r}_{\text{second-round effect}} \\
 &\quad - \underbrace{(C_y - TB_{im} - G_y)^2 \cdot (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot \Delta r}_{\text{third-round effect}} \\
 &\quad - \underbrace{(C_y - TB_{im} - G_y)^3 \cdot (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot \Delta r}_{\text{fourth-round effect}} + \dots \\
 &= - \left[\sum_{t=0}^{\infty} (C_y - TB_{im} - G_y)^t \right] \cdot (C_r + I_r + TB_\varepsilon \cdot \varepsilon_r) \cdot \Delta r \\
 &= \frac{-(C_r + I_r + TB_\varepsilon \cdot \varepsilon_r)}{1 - (C_y - TB_{im} - G_y)} \cdot \Delta r,
 \end{aligned}$$

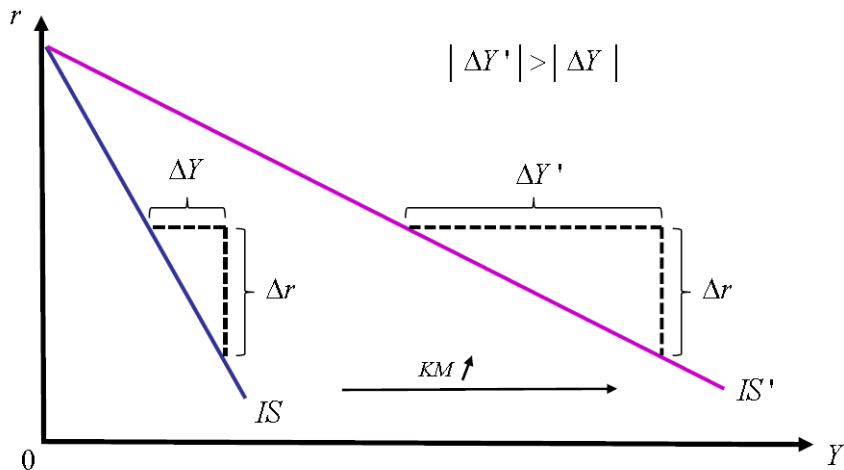
as predicted by (56), and with $\frac{1}{1 - (C_y - TB_{im} - G_y)}$ being the Keynesian multiplier in Equation (54).

As from Equations (52) and (54) the IS -curve is given by $r = IS_0^r - IS_1^r \cdot Y$, with

$$IS_r = -\frac{1 - (C_y - TB_{im} - G_y)}{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r} = -\left(\frac{1}{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}\right) \cdot \left(\frac{1}{KM}\right),$$

...

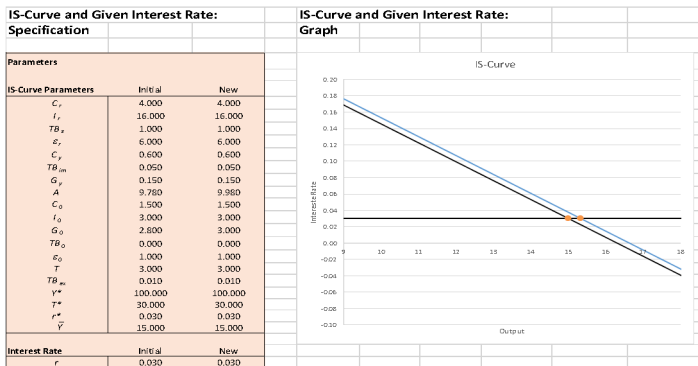
... the larger the Keynesian multiplier, the flatter the IS -curve, and the larger the output losses due to a given increase of the interest rate.



... the larger the Keynesian multiplier, the flatter the *IS*-curve, and the larger the output losses due to a given increase of the interest rate.

Workbook for the Keynesian Multiplier and Crowding Out Effects

The workbook *KeynesianMultiplierandCIFXChannels.xlsm* allows to study how changes in the autonomous components of aggregate demand and/or the interest rate affect output in the short-run macroeconomic outcome, ...



... the larger the Keynesian multiplier, the flatter the *IS*-curve, and the larger the output losses due to a given increase of the interest rate.

... how these output effects may be decomposed, and how the other endogenous variables change:

Keynesian Multiplier and Consumption, Investment as well as Exchange Rate Channels:		Initial and New Short-Run Macroeconomic Outcomes				
Quantitative Results						
Transmission Channels for Changes in Output Following Changes Exclusively in the Components of Autonomous Demand and/or in the Interest Rate		Initial and New Short-Run Macroeconomic Outcomes Following Changes in Any of the Model Parameters				
Keynesian Multiplier				Absolute Change	Percentage Change	
Stimulus: Change in A	0.200	Output	15.000	15.333	0.333	2.222
Size of Multiplier	1.667	Interest Rate	0.030	0.030	0.000	0.000
Change in Output	0.333	Consumption	8.580	8.780	0.200	2.331
		Investment	2.520	2.520	0.000	0.000
		Government Expenditure	2.800	2.950	0.150	5.357
		Trade Balance	1.100	1.083	-0.017	-1.515
		Exchange Rate	1.000	1.000	0.000	0.000
Interest Rate Change: Consumption Channel						
Change in Output	0.000					
Interest Rate Change: Investment Channel						
Change in Output	0.000					
Interest Rate Change: Exchange Rate Channel						
Change in Output	0.000					
Total Change in Output						
Sum of Effects	0.333					